

杨佳明

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教育背景

2025.08 – 2028.06　　**哈尔滨工业大学**　**（C9，保研）**　　控制科学与工程（硕士）

2020.09 – 2024.06　　**哈尔滨工业大学**　**（985，C9）GPA : 3.84 / 4.0**　　自动化（本科）

实习经历

2026.06 – 2026.09　上海赛力斯凤凰技术有限公司 – 数据部　　自动驾驶数据算法工程师

项目背景： 参与 **SACP 智驾数据云平台** 多传感器**预标注**模块建设，面向 **相机与激光雷达** 采集数据，构建“云端读取—检测与跟踪—后处理—质检—云端回写”的自动化数据生产链路，为人工精修及下游感知模型训练提供**动态、静态标注**数据。

工作内容：

- **动态标注 2D 质检功能：** 面向人工质检前的异常预筛，将相机 **2D 检测算法** 接入 LiDAR 3D 动态标注链路，完成 3D 框多相机投影、Ego ROI 过滤及 2D－3D 实例关联；基于**归一化重叠率**与**匈牙利算法**实现全局匹配，并构建疑似**误检、遮挡、视野外**三类异常分流机制，设计 3D 准确率和 2D 召回率任务级质量指标。在 8 个测试生产数据 CLIPs（3200 帧） 中完成 **6.46 万组实例匹配**，主要动态类别（Car、Truck、Bicycle、Pedestrian）**Precision、Recall 代理指标**分别达到 **75.0%**和 **84.5%**，自动标记 **1.21 万**个疑似误检 并补充 **1.18 万**个疑似漏检线索，辅助人工聚焦异常结果；
- **标注后处理链路：**封装**基于 CTRV-EKF 与 RTS 平滑**的**动态轨迹插值**、静态目标世界坐标**鲁棒聚合**算法，并接入动态标注链路，在测试用例中完成 **18.92 万/18.95 万**个候选 Feature 转换，处理覆盖率达 **99.86%**；开发静态总标注到逐帧 Ego 标注切帧功能，结合 **6DoF 坐标变换**、五路**点云融合**及 **KD-tree/OBB** 筛选生成逐帧真值标注静态目标，将静态目标人工标注工作量**压缩至原有的 1/44**；
- **云端生产接入：** 完成后处理与质检模块的可插拔集成，适配 **S3** 结果下载上传、**Meta** 写回、**Kafka** 状态上报以及 **Docker/Argo** 部署，补充异常降级、配置校验和自动化测试，开发功能已接入 SACP 云端生产流程进行**约 5.9w CLIPs** 的数据生产。

科研经历

2023.09 – 2024.06　《基于深度学习的无人机航拍图像目标检测研究》　　本科毕业设计

课题背景：无人机航拍视角下小目标检测与模型压缩。

- 构建包含 5 类目标、3224 张图像的无人机航拍数据集；
- 改进 YOLOv7 模型： 引入“EffectiveSE＋BatchNorm”注意力机制 ＋Network Slimming 通道剪枝，在保持 mAP 相近的前提下，模型参数量降低 50%，推理速度提升 13.4%；
- 研究成果发表于第 44 届中国控制会议（《Toward Optical Military Targets: Benchmark Dataset and Lightweight Detection Model》）。通过毕业答辩，最终成绩 92 分

2026.09 – 至今　　面向空地异构视角**的**几何一致三维高斯**联合重建　　硕士阶段研究方向**

课题背景：面向航拍与街拍图像的三维重建，研究跨视角几何关联与坐标系对齐，为后续 3D Gaussian Splatting 联合建模提供位姿和稀疏几何基础，现阶段聚焦 SfM 子模型重建与对齐。

- **跨视角对齐：** 构建**“检索—匹配—关联—对齐—评测”**五阶段流程。集成 **MASt3R** 检索与匹配，将二维对应关联至 SfM 三维点，结合 RANSAC 初值估计与双向重投影约束的七参数 Sim(3) 优化，实现空地坐标系对齐。在 Horizon-GS 数据集 Park 场景的 5,419 张地面图像对应的相机位姿上完成参考评测，归一化位置 RMSE 为参考尺度的 **0.286%**，旋转 RMSE 为 **0.197°**。
- **三维高斯重建：** 在自采场景复现 3DGS、2DGS，理解可微高斯光栅化的 CUDA 实现。

校园经历

2024.09 – 2025.07　新疆维吾尔自治区阿克苏地区库车市第三中学　　高一年级物理教师

- 获**“优秀青年教师”**、**“优秀支教教师”**荣誉称号，事迹被**中国青年网、人民日报**报道。

2025.09 – 至今　　哈尔滨工业大学航天学院 – 自动控制理论　　课程助教

技术栈

编程与开发　**C++**，**Python**；**Linux**，**Docker**，**Git**，CMake；PyTorch，OpenCV；**Codex**、Claude Code

办公软件　Visio，Mermaid，XMind

英语　CET-6，英语口语流利

Jiaming Yang



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Education

2025.08 – 2028.06 **Harbin Institute of Technology (C9; admitted by recommendation)**
Control Science and Engineering (Master's)

2020.09 – 2024.06 **Harbin Institute of Technology (Project 985, C9) GPA: 3.84 / 4.0**
Automation (Bachelor's)

Internship Experience

2026.06 – 2026.09 **Shanghai SERES Phoenix Technology Co., Ltd. – Data Department**
Autonomous Driving Data Algorithm Engineer

Project background: Contributed to the multi-sensor pre-annotation module of the SACP intelligent-driving data cloud platform. Built an automated data production pipeline for camera and LiDAR data, covering cloud retrieval, detection and tracking, post-processing, quality inspection, and cloud write-back, to provide dynamic and static annotations for manual refinement and downstream perception model training.

Responsibilities:

- **2D quality inspection for dynamic annotations:** Integrated camera 2D detection algorithms into the LiDAR 3D dynamic annotation pipeline to pre-screen anomalies before manual quality inspection. Implemented multi-camera projection of 3D boxes, ego-frame ROI filtering, and 2D–3D instance association; used normalized overlap ratios and the Hungarian algorithm for global matching, and routed anomalies into three categories: suspected false positives, occlusion, and out-of-view cases. Designed task-level quality metrics for 3D precision and 2D recall. Matched 64,600 instance pairs across 8 production-data test CLIPs (3,200 frames), achieving Precision and Recall proxy metrics of **75.0%** and **84.5%** for the main dynamic classes (Car, Truck, Bicycle, Pedestrian). Automatically flagged 12,100 suspected false positives and identified 11,800 potential missed detections to focus manual review on anomalous results;
- **Annotation post-processing pipeline:** Encapsulated dynamic trajectory interpolation using CTRV-EKF and RTS smoothing, together with robust aggregation of static objects in world coordinates, and integrated both algorithms into the dynamic annotation pipeline. Converted 189,200 / 189,500 candidate Features in test cases, achieving **99.86%** processing coverage. Developed frame-wise slicing of aggregated static annotations into ego-frame annotations, combining 6DoF coordinate transformations, five-stream point-cloud fusion, and KD-tree/OBB filtering to generate per-frame ground-truth static-object annotations, reducing the manual static-object annotation workload **to 1/44 of the original level**;
- **Cloud production integration:** Integrated post-processing and quality-inspection modules as pluggable components, supporting result download/upload via S3, Meta write-back, Kafka status reporting, and Docker/Argo deployment. Added exception fallbacks, configuration validation, and automated tests. The implemented features have been integrated into the SACP cloud production workflow to process **approximately 59,000 CLIPs**.

Research Experience

2023.09 – 2024.06 **Deep Learning-Based Object Detection in UAV Aerial Images**
Undergraduate Thesis

Research background: Small-object detection and model compression for UAV aerial imagery.

- Built a UAV aerial-image dataset comprising 5 target classes and 3,224 images;
- Improved YOLOv7 by introducing an “EffectiveSE + BatchNorm” attention mechanism and Network Slimming channel pruning, reducing model parameters by 50% and increasing inference speed by 13.4% while maintaining comparable mAP;
- Published the research at the 44th Chinese Control Conference (“Toward Optical Military Targets: Benchmark Dataset and Lightweight Detection Model”). Passed the thesis defense with a final grade of 92.

2026.09 – Present **Joint Reconstruction with Geometrically Consistent 3D Gaussians from Heterogeneous Aerial and Ground Views**
Master’s Research Focus

Research background: Investigating cross-view geometric association and coordinate-system alignment for 3D reconstruction from aerial and street-view images, providing camera poses and sparse geometry for subsequent joint 3D Gaussian Splatting modeling. The current focus is SfM submodel reconstruction and alignment.

- **Cross-view alignment:** Built a five-stage “retrieval–matching–association–alignment–evaluation” pipeline. Integrated MASt3R for retrieval and matching, associated 2D correspondences with SfM 3D points, and combined RANSAC initialization with seven-parameter Sim(3) optimization constrained by bidirectional reprojection to align aerial and ground coordinate systems. Evaluated the camera poses corresponding to 5,419 ground-view images against a reference in the Park scene of the Horizon-GS dataset, achieving a normalized positional RMSE of **0.286%** of the reference scale and a rotational RMSE of **0.197°**.
- **3D Gaussian reconstruction:** Reproduced 3DGS and 2DGS on self-captured scenes and developed an understanding of the CUDA implementation of differentiable Gaussian rasterization.

Campus Experience

2024.09 – 2025.07 **Kuqa No. 3 Middle School, Aksu Prefecture, Xinjiang Uyghur Autonomous Region**
Grade 10 Physics Teacher

- Received the “Outstanding Young Teacher” and “Outstanding Volunteer Teacher” honors; work featured by China Youth Network and People’s Daily.

2025.09 – Present **School of Astronautics, Harbin Institute of Technology – Automatic Control Theory**
Teaching Assistant

Technical Stack

Programming & Development	C++, Python; Linux, Docker, Git, CMake; PyTorch, OpenCV; Codex, Claude Code
Office Software	Visio, Mermaid, XMind
English	CET-6; fluent spoken English